Supervised Learning, COMP0078

Sample Examination Paper 3 (Level 2)

Suitable for Cohorts: 25/26

Answer ALL EIGHT questions.

Notation: Let $[m] := \{1, \ldots, m\}$. We also overload notation so that

$$[\text{pred}] := \begin{cases} 1 & \text{pred is true} \\ 0 & \text{pred is false} \end{cases}$$

Marks for each part of each question are indicated in square brackets.
Calculators are NOT permitted.

1.

(a) In least squares regression, we solve the normal equations $X^\top X\mathbf{w} = X^\top \mathbf{y}$. When is the solution unique? What problem arises when $X^\top X$ is singular, and how does ridge regression address this? [5 marks]

(b) Explain k-fold cross-validation. Why do we use cross-validation instead of simply minimizing training error to select hyperparameters? What is the tradeoff when choosing k? [5 marks]

2.

(a) Consider the feature map $\phi : \mathbb{R}^2 \to \mathbb{R}^3$ defined by $\phi(\mathbf{x}) = (x_1^2, \sqrt{2}x_1x_2, x_2^2)^\top$.

(i) Show that $\langle \phi(\mathbf{x}), \phi(\mathbf{z}) \rangle = (\mathbf{x}^\top \mathbf{z})^2$.

(ii) What kernel does this feature map correspond to?

[5 marks]

(b) The min kernel is defined as $k(x, y) = \min(x, y)$ for $x, y \geq 0$. Prove that this is a valid kernel by constructing an explicit feature map.

Hint: Consider indicator functions $\phi_t(x) = \mathbf{1}[x \geq t]$ for $t \in [0, \infty)$. [10 marks]

3.

(a) For kernel ridge regression, the primal solution requires inverting an $N \times N$ matrix (where N is the feature dimension), while the dual requires inverting an $m \times m$ matrix (where m is the number of samples).

When is the dual formulation computationally preferable? Give an example of a kernel where this matters. [5 marks]

(b) The soft-margin SVM can be written as:

$$\min_{\mathbf{w},b} \sum_{i=1}^{m} \max(0, 1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b)) + \frac{1}{2C}\|\mathbf{w}\|^2$$

Explain the role of the parameter C. What happens to the margin and training error as $C \to \infty$? As $C \to 0$? [5 marks]

4.

(a) For classification trees, three common node impurity measures are:

- Misclassification error: $1 - \max_k \hat{p}_k$
- Gini index: $\sum_k \hat{p}_k(1 - \hat{p}_k)$
- Cross-entropy: $-\sum_k \hat{p}_k \log \hat{p}_k$

where $\hat{p}_k$ is the proportion of class k in the node.

For a binary classification node with $\hat{p}_1 = 0.5$, compute all three measures. Which measures are more sensitive to changes in class probabilities near $\hat{p}_1 = 0.5$? [5 marks]

(b) AdaBoost can be derived as greedy minimization of the exponential loss $\sum_i \exp(-y_i f(\mathbf{x}_i))$.

Why is exponential loss preferred over squared loss $(y - f(\mathbf{x}))^2$ for classification? What is the connection between AdaBoost and the concept of "weak learners"? [5 marks]

5.

(a) The Winnow algorithm for learning monotone disjunctions updates weights multiplicatively:

$$w_{t+1,i} = w_{t,i} \cdot 2^{(y_t - \hat{y}_t)x_{t,i}}$$

Explain what happens to $w_{t,i}$ on a positive mistake ($y_t = 1, \hat{y}_t = 0$) versus a negative mistake ($y_t = 0, \hat{y}_t = 1$) when $x_{t,i} = 1$. [5 marks]

(b) A k-term DNF (Disjunctive Normal Form) is a disjunction of k terms, where each term is a conjunction of literals.

The ANOVA kernel $K(\mathbf{x}, \mathbf{y}) = \prod_{i=1}^{n}(1 + x_i y_i)$ can be used to learn DNF. What is the dimension of the feature space corresponding to this kernel? Why is this kernel useful despite the exponential feature dimension? [5 marks]

6.

(a) In spectral clustering, we want to minimize RatioCut:

$$\text{RatioCut}(A, B) = \text{cut}(A, B)\left(\frac{1}{|A|} + \frac{1}{|B|}\right)$$

This is NP-hard. Explain how the second eigenvector of the graph Laplacian provides a relaxed approximate solution. [5 marks]

(b) The effective resistance between vertices p and q in a graph is given by:

$$r_G(p, q) = L^+_{pp} + L^+_{qq} - 2L^+_{pq}$$

where L^+ is the pseudo-inverse of the Laplacian.

For the two-clique graph (two m-cliques connected by a single edge), explain qualitatively why the effective resistance between vertices in the same clique is much smaller than between vertices in different cliques. [5 marks]

7.

(a) In agnostic PAC learning, we remove the realizability assumption. The sample complexity for a finite hypothesis class becomes:

$$m \geq \frac{2\log(2|H|/\delta)}{\epsilon^2}$$

Compare this to the realizable case bound $m \geq \frac{\log(|H|/\delta)}{\epsilon}$. Why does the dependence on ϵ change from $1/\epsilon$ to $1/\epsilon^2$? [5 marks]

(b) The generalization error can be decomposed as:

$$L_{\mathcal{D}}(h_S) = \epsilon_{\text{app}} + \epsilon_{\text{est}}$$

where $\epsilon_{\text{app}} = \min_{h \in H} L_{\mathcal{D}}(h)$ and $\epsilon_{\text{est}} = L_{\mathcal{D}}(h_S) - \epsilon_{\text{app}}$.

Explain how increasing the complexity of H affects each term. What is the bias-complexity tradeoff? [5 marks]

8.

(a) Compare EXP3 (for bandit/partial feedback) to Hedge (for full information).

(i) What information does the learner receive after each round in each setting?

(ii) How does EXP3's regret bound compare to Hedge's?

[5 marks]

(b) Describe how to convert an online learning algorithm with mistake bound M into a batch learning algorithm. If the online algorithm has mistake bound M on any sequence, what can you say about the generalization error of the resulting batch algorithm? [5 marks]

END OF PAPER